

3 (a) Define specific latent heat of fusion.

[2]

(b) Explain why substances usually have a higher specific latent heat for vaporisation than for fusion.

..... [2]

(c) An ice cube of mass 15.6 g at a temperature of -18°C is removed from a freezer.

The ice cube is immediately placed in an insulated beaker that contains a mass of 132.4 g of water at a temperature of 34 °C.

The specific heat capacity of water is $4.19 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

The specific heat capacity of ice is $2.11 \text{ kJ kg}^{-1} \text{ } ^\circ\text{C}^{-1}$.

The specific latent heat of fusion of water is 336 kJ kg^{-1} .

The beaker has negligible specific heat capacity.

The contents of the beaker are left to reach thermal equilibrium.

Calculate the temperature, in $^{\circ}\text{C}$, of the water in the beaker when thermal equilibrium is reached.

temperature = °C [4]

[Total: 8]